Notes: Odean (American Economic Review, 1999)

Do Investors Trade Too Much?

Contents

1 Big picture

- **Question.** Do individual investors trade too much, in the sense that their trades lower expected returns after costs or even before costs?
- **Setting.** The paper studies 10,000 randomly selected discount brokerage accounts that were active in 1987 and follows their trades from January 1987 through December 1993.
- **Main empirical object.** Compare the future performance of stocks investors buy with the future performance of stocks they sell.
- **Core benchmark.** A rational informed investor who trades to improve performance should buy stocks that outperform the stocks he sells by at least the trading cost.
- **Trading cost.** The paper estimates a round-trip cost of about 5.9%, combining commissions on buys and sells plus an effective spread component.
- **Main result.** Investors' purchases do not outperform their sales. On average, the stocks bought underperform the stocks sold, even before subtracting trading costs.
- **Magnitude.** In the full sample, purchase-minus-sale raw return differences are -1.36%, -3.31%, and -3.32% over 84, 252, and 504 trading days.
- **Robustness.** The underperformance persists in cleaner replacement trades, active and inactive investor groups, early and late subperiods, calendar-time portfolio regressions, and market-adjusted returns.
- **Mechanism.** Precision-only overconfidence can explain trading that fails to cover costs, but it cannot by itself explain buying stocks that underperform the stocks sold. The evidence points to systematic misinterpretation of information.
- **Behavioral patterns.** Investors buy attention-grabbing winners and losers, sell recent winners disproportionately, and seem influenced by attention, the disposition effect, reluctance to short, and possibly mistimed momentum.
- **Economic takeaway.** Retail trading is not merely costly because commissions are high. The trades themselves are poor: investors tend to sell better-performing stocks and buy worse-performing stocks.

2 Data and measurement

2.1 Brokerage-account data

- The sample begins with 10,000 customer accounts randomly selected from accounts active in 1987 at a nationwide discount brokerage house.
- The trades file covers January 1987 to December 1993 and contains 162,948 records.
- The positions file contains monthly holdings from January 1988 through December 1993 and contains 1,258,135 records.
- Each trade record includes account identifier, trade date, security identifier, buy/sell indicator, quantity, commission, and principal amount.
- The analysis focuses on NYSE, American Stock Exchange, and NASDAQ common-stock trades with daily returns available in CRSP.
- The main CRSP-matched sample contains 97,483 trades: 49,948 purchases and 47,535 sales.

- Dollar trading is large for a retail sample: about \$530.7 million of purchases and \$579.9 million of sales.

Interpretation. The discount-brokerage setting is important because it minimizes broker churning and institutional agency explanations. The trades are mainly self-directed household decisions, so the sample is especially useful for testing behavioral explanations of excessive trading.

2.2 Trading costs and turnover

- Equal-weighted average commission on a purchase is 2.23% of purchase price.
- Equal-weighted average commission on a sale is 2.76% of sale price.
- The estimated effective spread component is about 0.94%.
- The average total round-trip cost is therefore about 5.9%.
- Average monthly turnover is 6.5%.
- Average size deciles of purchases and sales are very similar: 8.65 for purchases and 8.68 for sales, where 10 is the largest-size decile.

Interpretation. The 5.9% cost benchmark is large. A trade motivated by superior information must generate a very high expected spread between the bought and sold stocks merely to break even.

2.3 Outcome variables

- The paper computes average future returns for stocks bought and stocks sold.
- Main horizons are 84, 252, and 504 trading days after the transaction, approximately four months, one year, and two years.
- Returns are computed from CRSP daily return files.
- Return calculation begins the day after the transaction to avoid mechanically incorporating bid-ask bounce at the trade date.
- The key statistic is the future return on purchases minus the future return on sales.

Interpretation. The design asks a direct counterfactual question: after an investor switches from a sold stock into a bought stock, did the purchased stock perform better than the stock that was sold?

2.4 Partitions of the data

- **All trades:** the baseline sample of purchases and sales with CRSP returns.
- **Cleaner replacement trades:** purchases made within three weeks of profitable full-position sales, where the purchase has size decile no larger than the sale.
- **Trading intensity:** the 10% of investors who trade most versus the 90% who trade least.
- **Time periods:** 1987–1989 versus 1990–1993.
- **Return-pattern partitions:** previous winners versus previous losers, based on raw returns before the trade.

Interpretation. The cleaner replacement-trade sample is designed to remove alternative motives such as liquidity needs, tax-loss selling, rebalancing after appreciation, or deliberate movement into lower-risk stocks. The result becomes even stronger in this restricted sample.

3 Models and empirical specifications

3.1 Average post-trade return

For each transaction i, let j_i denote the security traded and t_i the transaction date. The average return to purchased securities over T trading days is

$$R_T^P = \frac{1}{N_P} \sum_{i=1}^{N_P} \left[\prod_{\tau=1}^{T} (1 + R_{j_i, t_i+\tau}) - 1 \right], \tag{1}$$

with an analogous expression for sold securities, R_T^S.

Interpretation. This is an event-time return calculation. It averages the future return path of every purchased stock and compares it with the future return path of every sold stock.

3.2 Null hypothesis N1: trading covers costs

$$H_0^{N1}: \quad E\left[R_T^P - R_T^S\right] \geq 5.9\%. \tag{2}$$

Interpretation. This is the rational-trading benchmark. If investors trade to increase returns, the stocks they buy should outperform the stocks they sell by at least the approximate round-trip trading cost. Rejecting this hypothesis means trading is not profitable net of costs.

3.3 Null hypothesis N2: purchases are at least as good as sales

$$H_0^{N2}: \quad E\left[R_T^P - R_T^S\right] \geq 0. \tag{3}$$

Interpretation. This is the stronger behavioral test. Rejecting N2 means that even before costs, investors buy stocks that perform worse than the stocks they sell. This result is too strong for a pure "overestimate precision but interpret information correctly" story.

3.4 Bootstrap empirical distribution

- Individual stock returns overlap across transactions, so conventional independent-observation tests are not appropriate.
- For each traded stock, the paper draws a replacement stock from the same size decile and book-to-market quintile.
- It keeps the original transaction dates and computes future returns for the replacement stocks.
- This procedure is repeated 1,000 times to form an empirical distribution of purchase-minus-sale returns under the null that buys and sells come from the same return distribution.

Interpretation. The bootstrap preserves important features of the trades, such as timing and stock characteristics, while asking whether the observed buy-sell return gap is unusually bad relative to comparable replacement securities.

3.5 Calendar-time portfolio tests

For a p-month formation window, define R_{Bpt} as the return on an equal-weighted calendar-time portfolio of recently bought stocks and R_{Spt} as the analogous portfolio of recently sold stocks. The CAPM specification is

$$R_{Bpt} - R_{Spt} = \alpha_p + \beta_p (R_{mt} - R_{ft}) + \varepsilon_{pt}. \tag{4}$$

The Fama-French three-factor specification is

$$R_{Bpt} - R_{Spt} = \alpha_p + \beta_p(R_{mt} - R_{ft}) + z_p SMB_t + h_p HML_t + \varepsilon_{pt}. \tag{5}$$

Interpretation. The intercept α_p measures whether the buy portfolio underperforms the sell portfolio after adjusting for broad market exposure and, in the three-factor model, size and book-to-market exposure.

3.6 Market-adjusted returns and market timing

$$R^{MA}_{j,t:t+T} = R_{j,t:t+T} - R^{VW}_{m,t:t+T}, \tag{6}$$

where R^{VW}_m is the CRSP value-weighted market return over the same interval.

Interpretation. Market-adjusted returns isolate stock selection by removing broad market movements. The paper also tests market timing by regressing market returns on prior-month order imbalance; the estimated timing coefficient is statistically insignificant.

4 Identification and empirical design

4.1 What makes the test credible

- The paper observes actual trades rather than survey intentions or laboratory choices.
- The core comparison uses stocks the same class of investors bought and sold, rather than comparing investors to a broad market benchmark only.
- Discount-brokerage accounts reduce broker-induced churning and delegated-manager agency concerns.
- The bootstrap matches replacement securities on size and book-to-market characteristics.
- Calendar-time tests address cross-sectional dependence in event returns.
- Market-adjusted returns and Fama-French tests address broad risk and style explanations.
- The cleaner replacement-trade sample filters out many non-speculative reasons for trading.

Interpretation. The evidence is persuasive because the same qualitative result appears under several designs: event-time returns, restricted trades, calendar-time portfolios, and market-adjusted returns all say that bought stocks underperform sold stocks.

4.2 What the paper cannot fully prove

- The paper cannot observe investors' beliefs directly; overconfidence and misinterpretation are inferred from behavior and outcomes.
- The sample comes from one discount brokerage, so external validity to all retail investors is not automatic.
- Some trades may still be motivated by unobserved liquidity, tax, risk, or portfolio constraints.
- Account attrition creates possible survivorship issues in later years.
- The paper does not estimate a structural model of learning, attention, or belief formation.

Interpretation. The paper shows that the trades are costly and badly chosen on average. It is more cautious about the exact psychological channel that causes the bad choices.

5 Tables and figures

5.1 Original tables and figures in the paper

- The paper contains three main empirical tables and six figures.
- The tables and figures below are directly cropped from the original PDF and grouped to save space.

5.2 Table 1: Average returns following purchases and sales

TABLE 1—AVERAGE RETURNS FOLLOWING PURCHASES AND SALES

Panel A: All Transactions

	n	84 trading days later	252 trading days later	504 trading days later
Purchases	49,948	1.83	5.69	−24.00
Sales	47,535	3.19	9.00	27.32
Difference		−1.36	−3.31	−3.32
N1		(0.001)	(0.001)	(0.001)
N2		(0.001)	(0.001)	(0.002)

Panel B: Purchases Within Three Weeks of Sales—Sales for Profit and of Total Position—Size Decile of Purchase Less Than or Equal to Size Decile of Sale

	n	84 trading days later	252 trading days later	504 trading days later
Purchases	7,503	0.11	5.45	22.31
Sales	5,331	2.62	11.27	31.22
Difference		−2.51	−5.82	−8.91
N1		(0.001)	(0.001)	(0.001)
N2		(0.002)	(0.003)	(0.019)

Panel C: The 10 Percent of Investors Who Trade the Most

	n	84 trading days later	252 trading days later	504 trading days later
Purchases	29,078	2.13	7.07	25.28
Sales	26,732	3.04	9.76	28.78
Difference		−0.91	−2.69	−3.50
N1		(0.001)	(0.001)	(0.001)
N2		(0.001)	(0.001)	(0.010)

Panels A-C.

Panel D: The 90 Percent of Investors Who Trade the Least

	n	84 trading days later	252 trading days later	504 trading days later
Purchases	20,870	1.43	3.73	22.18
Sales	20,803	3.39	8.01	25.44
Difference		−1.96	−4.28	−3.26
N1		(0.001)	(0.001)	(0.001)
N2		(0.001)	(0.001)	(0.001)

Panel E: 1987–1989

	n	84 trading days later	252 trading days later	504 trading days later
Purchases	25,256	0.05	1.47	20.44
Sales	26,732	1.70	4.88	22.95
Difference		−1.65	−3.41	−2.51
N1		(0.001)	(0.001)	(0.001)
N2		(0.001)	(0.001)	(0.006)

Panel F: 1990–1993

	n	84 trading days later	252 trading days later	504 trading days later
Purchases	29,078	4.67	12.29	32.04
Sales	26,732	5.93	16.44	38.89
Difference		−1.26	−4.15	−6.85
N1		(0.001)	(0.001)	(0.001)
N2		(0.004)	(0.001)	(0.005)

Notes: Average percent returns are calculated for the 84, 252, and 504 trading days following purchases and following sales in the data set trades file. Using a bootstrapped empirical distribution for the difference in returns following buys and following sells, the null hypotheses N1 and N2 can be rejected with p-values given in parentheses. N1 is the null hypothesis that the average returns to securities subsequent to their purchase is at least 5.9 percent greater than the average returns to securities subsequent to their sale. N2 is the null hypothesis that the average returns to securities subsequent to their purchase is greater than or equal to the average returns to securities subsequent to their sale.

Panels D-F and notes.

Figure 1: Original Table 1 directly cropped from Odean (1999), split into adjacent panels to improve readability.

Read:

- Panel A shows the headline result: purchases underperform sales by -1.36%, -3.31%, and -3.32% over 84, 252, and 504 trading days.
- Panel B is designed to remove liquidity, tax-loss, rebalancing, and risk-reduction motives; the underperformance becomes larger: -2.51%, -5.82%, and -8.91%.
- The 10% most active traders still buy stocks that underperform stocks they sell.
- The 90% less active traders show an even larger one-year gap: -4.28%.
- Both early and late sample periods reject N1 and N2 at the reported horizons.

Economic message. The table is the central evidence that trading is excessive. Investors do not merely fail to cover commissions; they tend to replace better-performing stocks with worse-performing stocks.

5.3 Tables 2 and 3: Calendar-time and market-adjusted robustness

TABLE 2—MONTHLY ABNORMAL RETURNS FOR CALENDAR-TIME PORTFOLIOS

Formation period	4 months	12 months	24 months
Panel A: Raw Returns			
Return	−0.293***	−0.225***	−0.137**
	(0.081)	(0.071)	(0.067)
Panel B: CAPM Intercept			
Excess return	−0.311***	−0.234***	−0.152**
	(0.080)	(0.073)	(0.068)
Beta	0.036**	−0.012	0.020
	(0.018)	(0.020)	(0.018)
Panel C: Fama-French Three-Factor Intercept			
Excess return	−0.249***	−0.207***	−0.136**
	(0.075)	(0.070)	(0.065)
Beta	−0.001	−0.007	0.008
	(0.019)	(0.020)	(0.019)
Size coefficient	0.031	0.075***	0.068***
	(0.028)	(0.026)	(0.024)
HML coefficient	−0.138***	−0.051	−0.025
	(0.035)	(0.032)	(0.029)

Notes: Raw returns (Panel A) are $R_{Bt} - R_{St}$, where R_{Bt} is the percent return in month t on a equally weighted portfolio with one position in a security for each occurrence of a purchase of that security by any investor in the data set in the 4, 12, or 24 months (the formation period) preceding month t and R_{St} is the percent return in month t on a equally weighted portfolio with one position in a security for each occurrence of a sale of that security by any investor in the data set in the 4, 12, or 24 months preceding month t. The CAPM intercept is estimated from a time-series regression of $R_{Bt} - R_{St}$ on the market excess return $R_{mt} - R_{ft}$. The Fama-French three-factor intercept is estimated from a time-series regressions of $R_{Bt} - R_{St}$ on the market excess return, a zero-investment size portfolio (SMB_t), and a zero-investment book-to-market portfolio (HML_t). Standard errors are in parentheses.

***,**Significant at the 1- and 5-percent level, respectively. The null hypothesis for beta (the coefficient estimate on the market excess return) is H_0: $\beta = 1$.

Table 2: Calendar-time portfolio abnormal returns.

TABLE 3—AVERAGE MARKET-ADJUSTED RETURNS FOLLOWING PURCHASES AND SALES

	n	84 trading days later	252 trading days later	504 trading days later
Purchases	49,948	−1.33	−2.68	−0.68
Sales	47,535	0.12	0.54	2.89
Difference		−1.45	−3.22	−3.57
N1		(0.001)	(0.001)	(0.001)
N2		(0.001)	(0.001)	(0.002)

Notes: Average percent returns in excess of the CRSP value-weighted index are calculated for the 84, 252, and 504 trading days following purchases and following sales in the data set trades file. Using a bootstarpped empirical distribution for the difference in market-adjusted returns following buys and following sells, the null hypotheses N1 and N2 can be rejected with p-values given in parentheses. N1 is the null hypothesis that the average market-adjusted returns to securities subsequent to their purchase is at least 5.9 percent greater than the average market-adjusted returns to securities subsequent to their sale. N2 is the null hypothesis that the average market-adjusted returns to securities subsequent to their purchase is greater than or equal to the average market-adjusted returns to securities subsequent to their sale. This table reports results for all investors over the entire sample period.

Table 3: Market-adjusted returns.

Figure 2: Original Tables 2 and 3 directly cropped from Odean (1999).

Read:

- Table 2 shows negative monthly buy-minus-sell returns for 4-, 12-, and 24-month portfolio formation periods.
- CAPM intercepts are -0.311%, -0.234%, and -0.152% per month.
- Fama-French three-factor intercepts are -0.249%, -0.207%, and -0.136% per month.
- The size and book-to-market coefficients suggest investors buy somewhat smaller and more growth-oriented stocks than they sell, but style adjustment does not remove the negative abnormal return.

- Table 3 shows that market-adjusted purchase-minus-sale differences remain negative: -1.45%, -3.22%, and -3.57%.

Economic message. The result is not driven by event-time dependence, broad market exposure, or simple style tilts. Purchased stocks underperform sold stocks even after calendar-time and market-adjusted robustness checks.

5.4 Figures 1 and 2: Before-and-after return paths for all investors and less active investors

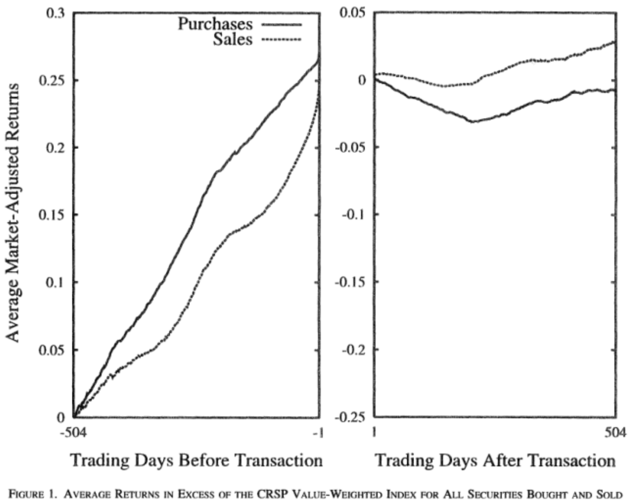

Figure 1: All purchases and sales.

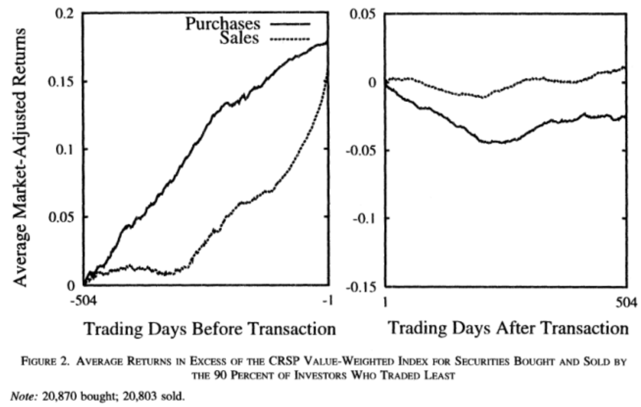

Figure 2: The 90% of investors who traded least.

Figure 3: Original Figures 1 and 2 directly cropped from Odean (1999).

Read:

- Before the transaction, both bought and sold stocks have generally outperformed the market.
- Investors tend to buy stocks with strong prior attention-grabbing performance.
- Stocks sold have stronger recent run-ups near the sale date.
- After purchase, bought stocks underperform sold stocks.
- The pattern is sharper among the less active 90% of investors, partly because their holding periods are longer and post-trade paths overlap less.

Economic message. The figures show the behavioral signature visually: investors are attracted to prior movers, but the stocks they buy subsequently do worse than the stocks they sell.

5.5 Figures 3 and 4: Previous winners and previous losers

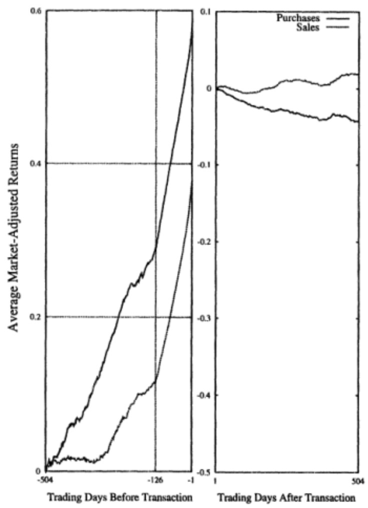

FIGURE 3. AVERAGE RETURNS IN EXCESS OF THE CRSP VALUE-WEIGHTED INDEX FOR SECURITIES BOUGHT AND SOLD BY THE 90 PERCENT OF INVESTORS WHO TRADED LEAST, FOR SECURITIES THAT HAD POSITIVE RAW RETURNS OVER THE 126 TRADING DAYS PRECEDING A PURCHASE OR A SALE

Figure 3: Stocks with positive raw returns before trade.

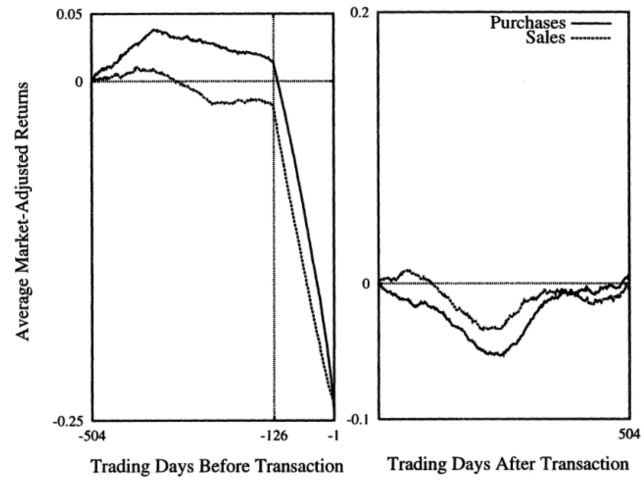

FIGURE 4. AVERAGE RETURNS IN EXCESS OF THE CRSP VALUE-WEIGHTED INDEX FOR SECURITIES BOUGHT AND SOLD BY THE 90 PERCENT OF INVESTORS WHO TRADED LEAST, FOR SECURITIES THAT HAD NEGATIVE RAW RETURNS OVER THE 126 TRADING DAYS PRECEDING A PURCHASE OR A SALE

Figure 4: Stocks with negative raw returns before trade.

Figure 4: Original Figures 3 and 4 directly cropped from Odean (1999).

Read:

- Figure 3 isolates previous winners. Bought winners had very large prior gains but then underperform; sold winners continue to perform better.
- Figure 4 isolates previous losers. Bought losers continue to underperform the market over the following year; sold losers do not look as bad over the same horizon.
- The figures show that poor purchase performance is not confined to either winners or losers.
- Investors sell far more previous winners than previous losers, consistent with the disposition effect.

Economic message. Investors appear to chase salient stocks on both sides: recent winners and dramatic losers. Their selling behavior is more shaped by realized-gain preferences, while their buying behavior reflects attention and possibly misread signals.

5.6 Figures 5 and 6: New issues and short-run pre-sale run-ups

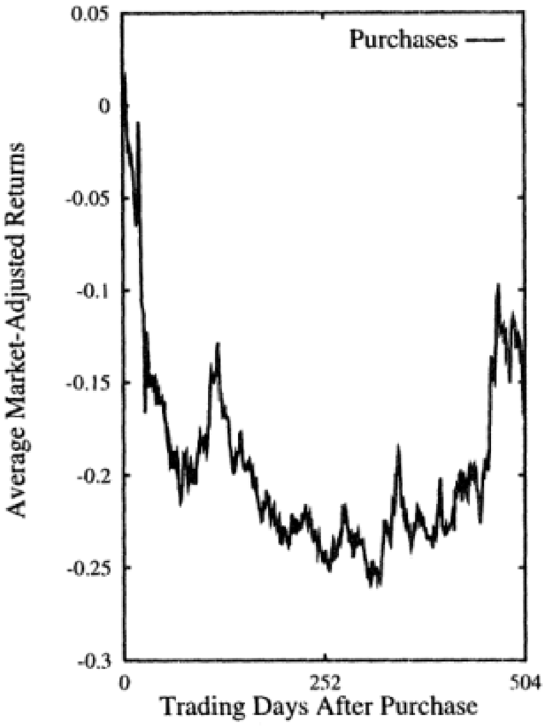

FIGURE 5. AVERAGE RETURNS IN EXCESS OF THE CRSP VALUE-WEIGHTED INDEX FOR SECURITIES BOUGHT THAT WERE ISSUED (LISTED ON CRSP) WITHIN FIVE DAYS PRIOR TO PURCHASE

Figure 5: Recently issued/listed stocks bought by investors.

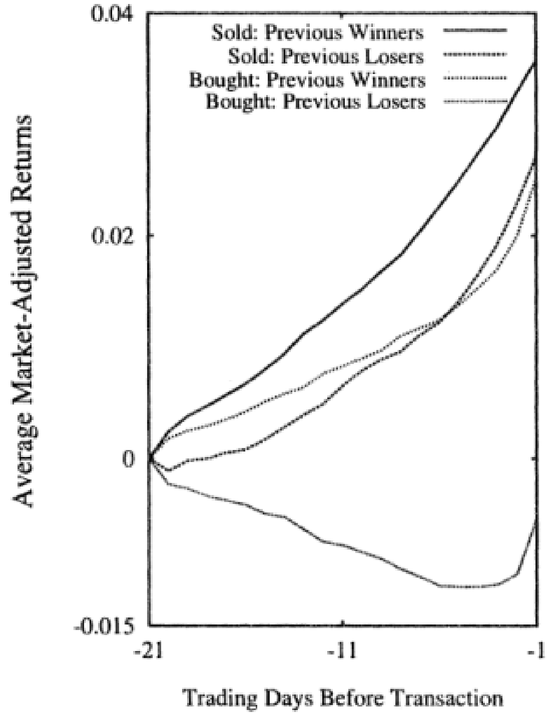

FIGURE 6. AVERAGE RETURNS IN EXCESS OF THE CRSP VALUE-WEIGHTED INDEX OVER THE 20 DAYS PRECEDING A TRANSACTION FOR SECURITIES BOUGHT AND SOLD BY THE 90 PERCENT OF INVESTORS WHO TRADED LEAST

Figure 6: The 20 trading days before transactions.

Figure 5: Original Figures 5 and 6 directly cropped from Odean (1999).

Read:

- Figure 5 uses purchases of securities listed on CRSP within five trading days as a proxy for new issues.
- These new issues underperform the market by roughly 25% over the first 14 months after purchase and then partially recover.
- Figure 6 shows sharp price increases in the 20 days before sales, both for previous winners and previous losers.
- Bought previous winners rise before purchase, while bought previous losers fall before purchase.
- The paper interprets these pre-trade patterns as consistent with attention, disposition effects, and moving reference points.

Economic message. Investors are pulled toward salient opportunities, including new issues and large price movers. On the sell side, recent gains can push stocks above a reference point and make investors willing to realize gains.

6 Short summary

- Odean studies whether individual investors trade excessively using account-level data from a discount brokerage.
- The main test compares the future returns of stocks bought with the future returns of stocks sold.
- A rational return-seeking trade should make the purchased stock outperform the sold stock by at least the estimated 5.9% round-trip cost.
- Instead, the stocks bought underperform the stocks sold, even before costs.
- The result holds at 84-, 252-, and 504-trading-day horizons.
- The result is stronger in a restricted sample designed to remove liquidity, tax-loss, rebalancing, and risk-reduction motives.
- Calendar-time portfolios and Fama-French regressions confirm that buy-minus-sell abnormal returns are negative.
- Market-adjusted tests show poor security selection rather than market timing as the central source of underperformance.
- Investors buy attention-grabbing stocks: previous winners, previous losers, and new issues.
- Investors sell recent gainers and sell more winners than losers, consistent with the disposition effect.
- The core behavioral interpretation is that investors are overconfident and, more importantly, systematically misinterpret the information they use to trade.

7 Conclusion

7.1 Core conclusion

Odean shows that discount-brokerage investors trade too much. Their trades lower returns because the stocks they buy perform worse than the stocks they sell, and this loss comes before adding commissions and spreads.

7.2 Economic intuition

A rational investor should trade only when expected gains exceed trading costs. Overconfident investors may overestimate the value of their information and trade too often. The surprising part of this paper is that investors do not merely overpay to trade; they trade in the wrong direction. This implies that many investors misread the signals they treat as informative.

7.3 Mechanism

The evidence points to a mix of overconfidence, attention, disposition effects, and reluctance to short. Investors have too many stocks to evaluate, so they focus on salient names. They buy stocks with extreme recent performance or news attention. They sell stocks after recent gains, especially winners, and hold losers too long. These choices create a systematic pattern in which purchased stocks underperform sold stocks.

7.4 Takeaway

Do Investors Trade Too Much? provides a clean empirical demonstration that individual investors' active trading is costly. The paper's deeper message is that the cost is not only mechanical transaction cost; it is also poor security selection driven by biased interpretation of information.